

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Applied Business Analytics Practical

Practical – IX

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Date of Assignment: 26/09/2026	Date/Time of Submission: 26/09/2026

Practical 9: Hypothesis Testing and A/B Testing

Part A: Hypothesis Testing

Dataset: supermarket_sales.csv

1. Load the supermarket_sales.csv dataset.
2. Display the first five records.
3. Select the revenue column.
4. Check for missing values.
5. Calculate the mean revenue.
6. Assume that the expected average revenue is ₹1,500. 7. Write the null hypothesis:

$$H_0: \mu = 1500$$

8. Write the alternative hypothesis:

$$H_1: \mu \neq 1500$$

9. Set the significance level as 5%.
10. Perform a one-sample t-test.
11. Calculate the p-value.
12. Compare the p-value with 0.05.
13. State whether the null hypothesis is rejected or not rejected. 14. Write the conclusion in simple business terms.

CODE:

```
import pandas as pd

import scipy.stats as stats

# Load dataset

df = pd.read_csv("supermarket_sales.csv")

# Show first 5 records

print(df.head())

# Use 'revenue' column for hypothesis testing

revenue = df['revenue']

# Check missing values

print("Missing values:", revenue.isnull().sum())

# Mean revenue

mean_revenue = revenue.mean()

print("Mean Revenue:", mean_revenue)

# Hypothesis setup

expected_mean = 1500

alpha = 0.05

# One-sample t-test

t_stat, p_value = stats.ttest_1samp(revenue, expected_mean)

print("T-statistic:", t_stat)

print("P-value:", p_value)

# Decision

if p_value < alpha:

    print("Reject Null Hypothesis")

    print("Conclusion: The average revenue is significantly
different from ₹1500.")
```

```
else:
```

```
    print("Fail to Reject Null Hypothesis")
```

```
    print("Conclusion: The average revenue is not significantly  
different from ₹1500.")
```

OUTPUT:

```
...      invoice_id branch      city customer_type gender_customer \
0  750-67-8428      A      Yangon      Member      Female
1  226-31-3081      C  Naypyitaw      Normal      Female
2  631-41-3108      A      Yangon      Normal      Male
3  123-19-1176      A      Yangon      Member      Male
4  373-73-7910      A      Yangon      Normal      Male

      product_line  unit_cost  quantity  5pct_markup  revenue \
0      Health and beauty      74.69          7      26.1415  548.9715
1  Electronic accessories      15.28          5       3.8200   80.2200
2      Home and lifestyle      46.33          7      16.2155  340.5255
3      Health and beauty      58.22          8      23.2880  489.0480
4      Sports and travel      86.31          7      30.2085  634.3785

      date  time payment_method  cogs  gm_pct  gross_income  rating
0  01/05/19 13:08      Ewallet  522.83  4.761905      26.1415      9.1
1  03/08/19 10:29        Cash   76.40  4.761905       3.8200      9.6
2  03/03/19 13:23  Credit card  324.31  4.761905      16.2155      7.4
3  1/27/2019 20:33      Ewallet  465.76  4.761905      23.2880      8.4
4  02/08/19 10:37      Ewallet  604.17  4.761905      30.2085      5.3
Missing values: 0
Mean Revenue: 322.966749
T-statistic: -151.37567896776733
P-value: 0.0
Reject Null Hypothesis
Conclusion: The average revenue is significantly different from ₹1500.
```

Part B: A/B Testing

Dataset: ab_testing.csv

The dataset contains:

Group	Users	Conversions
A	1000	120
B	1000	150

1. Load the A/B testing dataset.
2. Display the data.
3. Calculate the conversion rate for Group A.
4. Calculate the conversion rate for Group B.
5. Compare the conversion rates of A and B.
6. Write the null hypothesis:

$$H_0: p_A = p_B$$

7. Write the alternative hypothesis:

$$H_1: p_A \neq p_B$$

8. Set the significance level as 5%.
9. Perform a two-proportion test.
10. Calculate the p-value.
11. Compare the p-value with 0.05.
12. State whether the null hypothesis is rejected or not rejected.
13. Create a bar chart to compare the conversion rates.
14. Write the conclusion based on the test result.

CODE:

```
import numpy as np

from statsmodels.stats.proportion import proportions_ztest

import matplotlib.pyplot as plt

# A/B testing dataset

data = {

    'Group': ['A', 'B'],

    'Users': [1000, 1000],

    'Conversions': [120, 150]

}

# Display data

print(data)

# Conversion rates

conv_rate_A = data['Conversions'][0] / data['Users'][0]
```

```
conv_rate_B = data['Conversions'][1] / data['Users'][1]

print("Conversion Rate A:", conv_rate_A)
print("Conversion Rate B:", conv_rate_B)

# Hypothesis setup
alpha = 0.05

# Two-proportion z-test
successes = np.array([data['Conversions'][0], data['Conversions'][1]])
nobs = np.array([data['Users'][0], data['Users'][1]])

z_stat, p_value = proportions_ztest(successes, nobs, alternative='two-
sided')

print("Z-statistic:", z_stat)
print("P-value:", p_value)

# Decision
if p_value < alpha:

    print("Reject Null Hypothesis")

    print("Conclusion: Conversion rates between Group A and Group B are
significantly different.")
else:

    print("Fail to Reject Null Hypothesis")

    print("Conclusion: No significant difference in conversion rates
between Group A and Group B.")

# Bar chart
plt.bar(data['Group'], [conv_rate_A, conv_rate_B], color=['blue',
'green'])

plt.ylabel("Conversion Rate")

plt.title("A/B Testing Conversion Rates")

plt.show()
```

OUTPUT:

```
... {'Group': ['A', 'B'], 'Users': [1000, 1000], 'Conversions': [120, 150]}
Conversion Rate A: 0.12
Conversion Rate B: 0.15
Z-statistic: -1.963049807622355
P-value: 0.04964038686536878
Reject Null Hypothesis
Conclusion: Conversion rates between Group A and Group B are significantly different.
```

